

THE FEDERAL POLYTECHNIC, MUBI

(School of Science and Technology)

Department of Mathematics and Statistics

End of 1st and 3rd Semester Examinations 2021/2022 Session

Course Code: MTH 321

Course Title: Numerical Methods

Instruction: Answer any five (5) Questions

Time: 3 Hours

Question 1

Solve the system of equations by Gaussian elimination method.

$$2x + 3y - z = 5, 4x + 4y - 3z = 3 \text{ and } 2x - 3y + 2z = 2.$$

Question 2

Find by Gaussian elimination method, the inverse of A.

$$A = \begin{pmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{pmatrix}$$

Question 3

Form the divided difference table for the following data: -

x	-2	0	3	5	7	8
f(x)	-792	108	-72	48	-144	-252

Question 4

Use Langranges formular to fit a polynomial to the data.

x	-1	0	2	3
y	-8	5	1	12

Hence, find $y(1)$.

Question 5

(a) Find the first two derivatives of $(x)^{\frac{1}{3}}$ at; $x = 50$ and $x = 56$; given the table below.

x	50	51	52	53	54	55	56
$y = (x)^{\frac{1}{3}}$	3.6840	3.7084	3.7325	3.7563	3.7798	3.8030	3.8259

(b) Evaluate $\int_0^1 \frac{1}{1+x} dx$

Using (i) Trapezoidal rule

(ii) Simpson's rule

(iii) By actual integration

Question 6

Apply the fourth order Runge - Kutta method to find $y(0.2)$ given that $y' = x + y$, $y(0)=1$

Question 7

(a) Find an iterative formular to find the reciprocal of a given number, N and hence find the value of $\frac{1}{19}$.

(b) Solve $e^x - 3x = 0$, by the method of iteration.

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THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING TECHNOLOGY
END OF 1ST & 3RD SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION
SEPTEMBER, 2022.

PROGRAM: *HND II Electrical and Electronics Engineering Technology.*

UNIT: 2

COURSE TITLE: *Microprocessor Applications*

COURSE CODE: *EEE 438*

TIME ALLOWED: *2 ½ Hours*

INSTRUCTION: *Attempt any Four (4) Questions only. All question carries equal marks. Do not write on the question paper*

QUESTION ONE

- a) _____ to be processed by the digital system are made up of decimal digits, alphabetic characters, and special characters. (2½ marks)
- b) In the _____ system, the radix is 2 and the two allowed digits are 0 and 1. (2½ marks)
- c) The total number in a digit is called _____. (2½ marks)
- d) _____ is a graphical representation of a logic functions truth table. (2½ marks)
- e) If each position of a number does not represent a specific weight, then the coding scheme is called _____. (2½ marks)
- f) Digital applications can in general be classified into _____ types. (2½ marks)
- g) The name _____ come from the fact that these devices operate like a gate to let things (in this case, current) through. (2½ marks)
- h) _____ logic circuits are built round combinations of AND, OR and Invert circuits. (2½ marks)
- i) _____ are logic circuits whose outputs are a function of the present states of the inputs but also of its previous state. (2½ marks)
- j) _____ is the most basic building block in a design. (2½ marks)

QUESTION TWO

- a) In _____, a bus is a special kind of signal that may have its drivers turned off. (2½ marks)
- b) _____ is VHDL's term for a parameter that passes information to an entity. (2½ marks)
- c) A _____ is one which gives the output state of a logic function for all possible input combination. (2½ marks)
- d) The practically constructible logic functions associative or commutative property that have the ability to implement Boolean functions alone or in conjunction with other gates is called _____. (2½ marks)
- e) A graph with nodes and directed edges connecting the nodes that portrays the operation of the flip-flop is called _____. (2½ marks)
- f) The extra inputs of flip-flop are called _____ because they can set or reset the flip-flop regardless of the status of the clock signal. (2½ marks)
- g) The number of standards loads the output of a gate can drive without impairing its normal operation is called _____. (2½ marks)

- h) _____ are defined by the basic electronic (a NAND or a NOR gate) upon which complex digital circuits and functions are developed. (2½ marks)
- i) _____ is a basic unit of execution in VHDL. (2½ marks)
- j) Think of a _____ as a toolbox that contains tools used to build designs. (2½ marks)

QUESTION THREE

- a) Define System Software and briefly explain the types of System software. (13 marks)
- b) How does an operating system manage the memory of a computer? (6 marks)
- c) Explain briefly:
- Multiprogramming. (3 marks)
 - Multitasking. (3 marks)

QUESTION FOUR

- a) Differentiate between Hardware interface and Software interface. (6 marks)
- b) Using clear flow chart diagram only, show the interrupt driven I/O processing cycle. (11 marks)
- c) How does the DMA controller I/O work? (8 marks)

QUESTION FIVE

- a) Trace the IC evolution from SSI to VLSI. (8 marks)
- b) Briefly explain the following:
- RISC. (5½ marks)
 - CISC. (5½ marks)
- c) Define the following:
- Instruction Set. (2 marks)
 - Bandwidth. (2 marks)
 - Clock speed. (2 marks)

QUESTION SIX

- a) Compare between 8085 and 8086 microprocessors. (12 marks)
- b) List and briefly explain the components of a Digital Signal Processor (DSP). (6 marks)
- c) Explain briefly the working sequence of microprocessor. (7 marks)

THE FEDERAL POLYTECHNIC, MUBI
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING TECHNOLOGY
END OF 1ST & 3RD SEMESTER EXAMINATION 2021/2022 ACADEMIC SESSION
SEPTEMBER, 2022.

PROGRAMME: *HND II Electrical and Electronics Engineering Technology*
COURSE TITLE: *Electrical Measurement and Instrumentation IV*
COURSE CODE: *EEI 431*

UNITS: 2

TIME ALLOWED: 2½ Hours

INSTRUCTIONS: Answer any four (4) questions. All questions carry equal mark. Do not write on the question paper

QUESTION ONE

- (a) Briefly explain with the aid of diagram the principle of Murray loop test for earth fault test. (12mks)
- (b) In locating a ground fault by Varley loop test the following observations were recorded:
- (i) With key at stud 1, a balance was obtained when $P=10$, $Q=80$ and $S=2,400$
 - (ii) With key at stud 2, a balance was obtained when $P=10$, $Q=80$ and $S=3,400$
- Determine the distance of the fault from the testing end. Each conductor is 5km long. (8mks)

QUESTION TWO

- (a) Draw the block diagram of a strip chart recorder and briefly explain its principles of operation. (10mks)
- (b) With the aid of a suitable diagram describe the principles of operation of an X-Y recorder. (10mks)

QUESTION THREE

- (a) Briefly explain the following types of transducers:
- (i) An inductive transducer operating on the principle of variation of self-inductance for linear displacement measurement. (6mks)
 - (ii) A capacitive transducer operating on the principle of variation of distance between plates. (6mks)
- (b) An LVDT is used for measuring the deflection of a bellows. The sensitivity of LVDT is 40 V per mm. The bellows is deflected by 0.125 mm by pressure of 0.8×10^6 N per m^2 and the pressure when the voltage output of LVDT is 3.5V. (8mks)

QUESTION FOUR

- (a) With the aid of a circuit diagram briefly describe the principles of operation of a Linear Voltage Differential Transformer (LVDT) recorder. (10mks)
- (b) Use appropriate diagram, briefly explain the working principle of a bridge type null recorder. (10mks)

QUESTION FIVE

- (a) Briefly describe with the aid of block diagram the operating principle of a chopper type D.C voltmeter. (8mks)
- (b) Briefly explain with the aid of a neat block diagram the operating principle of a successive-approximation digital voltmeter. (12mks)

QUESTION SIX

- (a) State five (5) applications of thermistors and three (3) applications of thermocouples. (8mks)

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END OF 1ST AND 3RD SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION
SEPTEMBER, 2022.

PROGRAM: *HND II Electrical and Electronics Engineering Technology*

UNITS: 3

COURSE CODE: EEC 434

COURSE TITLE: *Analogue Electronics IV*

TIME ALLOWED: 3 Hours

INSTRUCTIONS: *Attempt Any Five (5) Questions. All Questions Carry Equal Marks. Do not write on the question paper*

QUESTION ONE

- a. Sketch a well labeled block diagram of an amplifier with a feedback path. [10marks]
- b. With reference to the diagram of 1a above, derive the formula for an amplifier gain with a negative feedback path (A^f). [10marks]

QUESTION TWO

In a tabular form, briefly identify any five (5) major comparisons between positive feedback and negative feedback of an amplifier. [20marks]

QUESTION THREE

- a. List the types of multi-vibrators. [6marks]
- b. From the listed multi-vibrators in 3a above, briefly explain one. [6marks]
- c. Explain briefly the following: (i) Clock Frequency (ii) Clock Period (iii) Clock Width (iv) Duty Cycle [2marks Each]

QUESTION FOUR

- a. Draw a comprehensive circuit diagram of a D.C power supply (Full Wave Bridge Rectifier) [14marks]
- b. List any three (3) advantages of a full wave rectifier. [6marks]

QUESTION FIVE

- a. Define an Oscillator. [5marks]
- b. With the aid of a circuit diagram, briefly explain the principles of operation of Hartley Oscillator using Op Amp. [10marks]
- c. State five (5) type of Oscillator. [5marks]

QUESTION SIX

- a. With the aid of a circuit diagram, briefly explain L.C Oscillator tank Circuit. [10mark]
- b. State five (5) major effects of applying negative feedback to an amplifier. [10marks]

QUESTION SEVEN

- a. In a tabular form, compare Harmonic Oscillator and Relaxation Oscillator. [18marks]
- b. An amplifier has an open-loop gain of 35. What is its voltage gain with 10% feedback? [2marks]

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END OF 1ST AND 3RD SEMESTER EXAMINATION 2021/2022 ACADEMIC SESSION
SEPTEMBER, 2022.

UNIT: 2

PROGRAMME: HND II Electrical and Electronics Engineering

COURSE TITLE: Electronic Design and Drafting

COURSE CODE: EEC437

TIME ALLOWED: 2½ HOURS

INSTRUCTIONS: Attempt any four (4) questions. All questions carry equal marks. Do not write on the question paper.

QUESTION ONE

- (a) Draw clearly the symbols and briefly state the functions of the following commonly used electronic components;
- (i). Preset resistor. (ii) Preset capacitor. (iii) Photo diode. (iv) Potentiometer. (v) Thyristor. (vi) Zener Diode. (vii) NAND gate. (viii) NOR gate. (ix) Reversing switch. (x) Piezo Transducer. *(1mk each)*
- (b) Identify the following passive components with their color codes where applicable;
- (i) 10MΩ. (ii) 2K2. (iii) 223K. (iv) 104C (v) 102 (vi) 4K7 *(½ mk each)*
(2mk each)
- (c) (i) List the three (3) basic standard transistor marking schemes
(ii) Briefly state and identify active components using letters by indicating the meaning of the first and second letters of the required device. *(6mks)*

QUESTION TWO

- (a) With the aid of neat schematic diagram of a signal tracer. *(4mks)*
- (i) Briefly explain a schematic diagram. *(6mks)*
- (ii) State the two standards for schematic symbols. *(2mks)*
- (iii) Outline the basic guideline for drawing a well-arranged schematic diagram. *(4mks)*
- (b) Differentiate between logic and block diagram. *(6mks)*
- (c) State three (3) reasons why block diagrams are important. *(3mks)*

QUESTION THREE

- (a) State and explain briefly, three (3) types of boards commercially available for building your project. *(4mks each)*
- (b) Outline the step-by-step procedure in laying out and building a circuit neatly on a strip board. *(6mks)*
- (c) State four (4) advantages and three disadvantages of printed circuit board over point-to-point or interconnection using wire. *(7mks)*

QUESTION FOUR

- (a) Briefly describe the following processes of producing a printed circuit board;
- (i). Print and etch method. *(4mk)*
- (ii) Photo-reduction method. *(4mk)*
- (iii) Any other method you know. *(4mk)*
- (b) Outline the basic stages in producing a PCB *(4mk)*
- (c) Briefly explain the following types of PCB assembly;
- (i) Single sided type. *(2mk)*
- (ii) Double sided type. *(2mk)*
- (iii) Multilayer type *(2mk)*
- (d) State three (3) advantages of fibre glass over sheet resin bonded paper (SRBP) board. *(3mks)*

QUESTION FIVE

- (a) Use neat figures to briefly explain the following diagrams
- Point to Point.
 - Baseline.
- (b) List the typical manufacturers specifications for the following devices.
- Audio amplifier.
 - Audio mixer.
 - Microphone.

(4mks)

(4mks)

(3mk)

(3mk)

(3mk)

- (c) Design a closed-circuit television system for the school of engineering complex to enable the dean monitor activities outside the complex via three cameras, three microphones, a video mixer, audio mixer, an amplifier and a loudspeaker.

(8mks)

QUESTION SIX

- (a) Differentiate between highway and baseline wiring diagrams.

(8mks)

- (b) Explain briefly manufacturers' specification meant for a device.

(4mks)

- (c) Design and draft a two-input microphone PA system for use where mains supplied voltage is 110V, for a client who requires 500 Watts of audio power. Use available units given in Table Q7c.

(13mks)

Table Q7c

Units	A	B	C	E
Microphone	Impedance-600Ω Cost = #1,200	Impedance-600Ω Cost = #1,300	Impedance-800Ω Cost = #1,000	Impedance-300Ω Cost = #550
Mixer	Input impedance = 600Ω Output Impedance = 600Ω No of inputs = 2 Voltage = 220v Cost = #4,200	Input impedance = 800Ω Output Impedance = 600Ω No of Inputs = 3 Voltage=110v Cost = #6,200	Input Impedance = 300Ω Output Impedance = 600Ω No of inputs = 3 Voltage=110v Cost = #5,200	Input impedance = 800Ω Output Impedance = 600Ω No of inputs = 2 Voltage = 220v Cost = #5,200
Amplifier	P out-500W Input impedance-600Ω Output impedance-8Ω Voltage 110v Cost = #5,000	P out-700W Input impedance-600Ω Output impedance-8Ω Voltage=110v Cost = #6,200	P out-500W Input impedance-600Ω Output impedance-8Ω Voltage = 220v Cost = #5,900	P out-550W Input impedance-500Ω Out impedance-4Ω Voltage = 220v Cost = #6,500
loudspeaker	Power-250W Impedance-16Ω Cost = #1,000	Power-300W Impedance-8Ω Cost = #2,500	Power-500W Impedance-8Ω Cost = #5,000	Power-500W Impedance-4Ω Cost = #2,200

THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL & ELECTRONIC ENGINEERING
1ST AND 3RD LONG VACATION REPEAT PROGRAMM (LVRP) SEMESTER EXAMINATION, 2020/2021
SESSION ACADEMIC SESSION
FEBRUARY, 2022.

PROGRAMME: IIND II
COURSE TITLE: Digital Communication II
COURSE CODE: EEE435
TIME ALLOWED: 2½ Hours
INSTRUCTIONS: Answer any five (5) questions. All questions carry equal marks. Do not write on question paper.

QUESTION ONE

- a) Define the following
 - i) A pulse signal (5mrks)
 - ii) A pulse code modulation signal (5mrks)
- b) Define the following concept
 - i) Quantization ii) Sampling iii) modulation (5mrks)
- c) Sketch the diagram of the PCM communication system and briefly explain the difference between Sampling and quantization (5mrks)
- d) Explain briefly the differential quantizing scheme (3mrks)

QUESTION TWO

- a) (i) State six advantages of the PCM system over analog modulation schemes. (3mrks)
- (ii) With the aid of diagrams explain briefly the delta modulation scheme, with its merits. (3mrks)
- b) Explain briefly with the aid of a diagram the two difficulties associated with delta modulation scheme. (5mrks)
- c) (i) Explain briefly the concept Multiplexing. (2mrks)
- (ii) State and explain briefly the two types of Multiplexing scheme. (5mrks)
- (iii) State 4 reasons why impedance matching is employed in coupling network on antenna. (2mrks)

QUESTION THREE

- a) i) Define the FM and AM systems. (2mrks)
- ii) With the aid of a block diagram explain briefly the principle of operation of an AM transmitter. (4mrks)
- b) i) Differentiate between an oscilloscope and a spectrum analyzer. (2mrks)
- ii) With the aid of a diagram, discuss briefly the direct π -coupling networks that are employed in a transmission systems. (4mrks)
- c) i) Briefly explain the following with respect to radio receivers. (4mrks)
- i) Selectivity (4mrks)
 - ii) Sensitivity
- ii) State four major differences between AM and FM receivers.

QUESTION FOUR

- a) i) With the aid of a diagram, sketch the essential features of the Tuned Radio Frequency (TRF) and the Superheterodyne Radio receivers. State the differences in their features. (7mrks)
- ii) State the function and the advantages of the RF amplifier in the super heterodyne radio. (3mrks)
- b) In a broadcast super heterodyne receiver having no RF amplifier, the loaded Q of the antenna coupling circuit (at the input to the mixer) is 100. If the intermediate frequency is 455kHz, calculate
 - i) The image frequency and its rejection ratio at 1000kHz (10mrks)
 - ii) The image frequency and its rejection ratio at 25MHz

QUESTION FIVE

- a) i) The function of the oscilloscope may be defined as capturing, displaying and analyzing a time domain waveform. Discuss the component blocks that make this possible. (5mrks)
ii) Define image frequency rejection in terms of broadcast band radio (2mrks)
- b) i) Describe the basic Telephone system (4mrks)
ii) Explain the concept of Telephone automatic dialing switching method (4mrks)
- c) Describe the principle of operation of a "Telefax" machine (5mrks)

QUESTION SIX

- a) (i) Draw the block diagram of the monochrome TV transmitter and state the purpose of the camera tube in the system. (10mrks)
(ii) Briefly explain the term "Persistence of vision" with respect to the human eye (4mrks)
- b) State three conditions that must be satisfied in color TV in order to be compatible (6mrks)

QUESTION SEVEN

- a) (i) Define the Y, I and Q signals as applied to colour TV system and State their relationship with the primary colours mathematically (7½mrks)
(ii) Draw the color triangle and indicate its seven (7) components. (3½mrks)
- b) (i) Compare the American and European TV systems in terms of the following: Number of lines per frame; Number of frame per seconds; field frequency; Line frequency and Video bandwidth in MHz. (5mrks)
- (ii) Define the acronym "GSM" with regards to telephony and draw a ten (10) cell circular structure of an area covered by the GSM network (4mrks)

FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING TECHNOLOGY
END OF 1ST AND 3RD SEMESTER EXAMINATION 2021/2022 SESSION
SEPTEMBER, 2022

PROGRAMME: *HND II Electrical and Electronics Engineering Technology*

UNIT: 3

COURSE TITLE: *Digital Communication II*

COURSE CODE: *EEE 435*

TIME ALLOWED: *3 hours*

INSTRUCTION: *Attempt any five (5) questions. All question carries equal marks. Do not write on the question paper*

QUESTION ONE

- a) Define the following:
 - i) Data Communication (5mrks)
 - ii) A pulse code modulation signal. (5mrks)
- b) What are the three (3) conditions for effective communications.
- c) Sketch the diagram of the PCM communication system and explain the difference between Sampling and quantization. (5mrks)
- d) Explain the Concept "differential quantizing scheme" (5mrks)

QUESTION TWO

- a) With the aid of a diagram describe the Elements of Communication system. (4mrks)
- b) Explain briefly the conditions for effective Communication system. (3mrks)
- c) Define the concept Multiplexing. (3mrks)
- d) State and explain the two types of Multiplexing scheme. (5mrks)
- e) State Five advantages of the PCM system over analog modulation schemes. (5mrks)

QUESTION THREE

- a) i) Define the AM and FM systems. (2mrks)
- ii) With the aid of a block diagram explain the principle of operation of an AM transmitter. (4mrks)
- b) i) Differentiate between an oscilloscope and a spectrum analyzer. (2mrks)
- ii) Assume we need to download text documents at the rate of 100pages per minute.
What is the required bit rate of the channel? (Assuming each page is an average of 24 Line with 80 characters in each line and one character requires 8 bits) (4mrks)
- c) i) Explain the following with respect to radio receivers
 - i) Selectivity. (4mrks)
 - ii) Sensitivity. (4mrks)
- ii) State four major differences between AM and FM receivers. (4mrks)

QUESTION FOUR

- a) i) With the aid of a diagram, sketch the essential features of the Tuned Radio Frequency (TRF) and the Super heterodyne Radio receivers. State the differences in their features. (7mrks)
- ii) State the function and the advantages of the RF amplifier in the Superheterodyne radio. (3mrks)
- b) In a broadcast superheterodyne receiver having no RF amplifier, the loaded Q of the antenna coupling circuit (at the input to the mixer) is 100. If the intermediate frequency is 455kHz, calculate
 - i) The image frequency and its rejection ratio at 1000kHz. (5mrks)
 - ii) The image frequency and its rejection ratio at 25MHz. (5mrks)

QUESTION FIVE

- a) i) The function of the oscilloscope may be defined as capturing, displaying and analyzing a time domain waveform. Discuss the component blocks that make this possible. (5mrks)
- ii) Define image frequency rejection in terms of broadcast band radio. (2mrks)
- b) i) Describe the basic Telephone system. (4mrks)
- ii) Explain the concept of Telephone automatic dialing switching method. (4mrks)
- c) Describe the principle of operation of a "Telephone" (5mrks)

QUESTION SIX

- a) (i) Draw the block diagram of the monochrome TV transmitter and state the purpose of the camera tube.
(10mrks)
(ii) Briefly explain the term "Persistence of vision" with respect to the human eye. (4mrks)
b) State three conditions that must be satisfied in color TV in order to be compatible. (6mrks)

QUESTION SEVEN

- a) (i) Define the Y, I and Q signals as applied to colour TV system. State their relationship with the primary colours mathematically. (6mrks)
(ii) Draw the color triangle and indicate its seven (7) components. (4mrks)
b) (i) Compare the American and European TV systems in terms of the following: Number of lines per frame; Number of frames per seconds; field frequency; Line frequency and Video bandwidth in MHz (5mrks)
(ii) Define synchronization with respect to television. (5mrks)

THE FEDERAL POLYTECHNIC, MUBI
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END OF 1ST AND 3RD SEMESTER EXAMINATION 2021/2022 SESSION
SEPTEMBER, 2021.

PROGRAMME: HND II Electrical and Electronics Engineering Technology
COURSE TITLE: Control Engineering III
COURSE CODE: EEC-433
TIME ALLOWED: 3 hours

UNIT: 3

INSTRUCTIONS: Attempt any five (5) questions only. All questions carry equal marks. Do not write on the question paper

QUESTION ONE

- a) Write down the speed equation of a d.c motor defining every quantity in the equation. (8 marks)
- b) With the aid of a diagram briefly explain the principles of operation of armature speed control. (8 marks)
- c) What are the advantages and disadvantages of field speed control over armature control. (4marks)

QUESTION TWO

- (a) With the aid of neat diagram briefly explain Ward-Leonard speed control mechanisms. (8 marks)
- (b) Briefly explain the applications of Ward-Leonard speed control mechanism. (6marks)
- (c) A 500V dc shunt motor takes 4A on no-load and runs at 750rpm. The resistance of armature and field windings are 0.2 ohms and 110 ohms respectively. Calculate the speed when the motor is loaded and taking a current of 50A. Assume the armature reaction weakens the field by 3%. (6marks)

QUESTION THREE

- (a) Briefly explain the performance of speed control system; (3marks)
 - i. Without negative feedback. (3marks)
 - ii. With negative feedback.
- (b) Briefly explain with the aid of a schematic diagram the principle of operation of Velodyne speed control system. (7marks)
- (c) Briefly explain with neat sketch a d.c motor speed control using a single-phase thyristor bridge rectifier speed control system. (7marks)

QUESTION FOUR

- (a) Briefly explain with the aid of a diagram potentiometer based basic position control system (10marks)
- (b) Briefly explain the following damping methods associated with speed/position control; (5marks)
 - i. Negative velocity feedback. (5marks)
 - ii. Positive acceleration feedback.

QUESTION FIVE

- (a) with the aid of suitable diagram briefly explain the automatic temperature control process. (7marks)
- (b) Briefly explain the following terms associated with process control; (2marks)
 - i. Set-point. (2marks)
 - ii. Deviation (Off set). (2marks)
 - iii. Distance-velocity lag.
- (c) crude oil is transported through a pipe of diameter 61 cm from Niger Delta to Kaduna refinery at the velocity of 100km/hour. If the distance from Niger Delta to Kaduna refinery is 100km. determine, (5marks)
 - i. The flow rate in gallons / hour. (2marks)
 - ii. Distance velocity lag.

QUESTION SIX

Illustrate with the associated symbols the implementation of the following operations in analogue computer:

- a) Add: $a + b$
- b) Subtract: $x - y$

(3 marks)

- c) Integrate: $\frac{d^2x}{dt^2}$

(3marks)

- d) Differentiate: y

(4marks)

- e) Realize: $2ay$

(2marks)

- f) Realize: $\frac{x}{2} + \frac{y}{2} + \frac{z}{2}$

(2marks)

- g) Integrate in summation form; $x_1 + x_2x_3$

(3marks)

(3marks)

QUESTION SEVEN

Draw the analogue computer flow diagrams to simulate:

- a) $2x + 3y = 5$, solve for x and draw the flow diagram
- b) $3x + 3y = 5 \dots (1)$
 $5x + 4y + 5 \dots (3)$ solve for x and y , then draw the flow diagram

(5marks)

(9marks)

(3marks)

(3marks)

- (c) Briefly explain the following;

- i. Amplitude scaling
- ii. Time scaling

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END OF 1st & 3rd SEMESTER EXAMINATION, 2021/2022 SESSION

SEPTEMBER, 2022
PROGRAMME HND II Electrical and Electronics Engineering
COURSE TITLE: ELECTROMAGNETIC FIELD THEORY
COURSE CODE: EEC 431
TIME ALLOWED: 2HRS 30MINS
INSTRUCTION: ANSWER ANY FOUR QUESTION. ALL QUESTION CARRY EQUAL MARKS

UNITS 2

QUESTION ONE

- Define Electromagnetic field theory
- Mention five Electromagnetic devices.
- Express Vector

(5marks)

(5marks)

$$B = \frac{10}{r} a_r + r \cos \theta a_\theta + a_\phi \text{ in Cartesian}$$

Find B (-3.4, 0)

(15Marks)

QUESTION TWO

- Mention five applications of Electrostatic in Agriculture
- State Coulomb's Law
- Two points charges $-4\mu\text{C}$ and $5\mu\text{C}$ are located at (2, -1, 3) and (0, 4, -2) respectively.
 find (i) The potential V_p at P(1, 0, 1)
 (ii) if point charge $3\mu\text{C}$ is located at the origin in addition to the two charges above, find the potential at (-1, 5, 2) assuming zero potential at infinity.

(5marks)

(8marks)

(6marks)

(6marks)

(6marks)

QUESTION THREE

- State Biot-Savart's law.
 ii. Deduced (ai) above in vector form.
- The y- and z-axes respectively of a semi-infinite conductor carry filamentary currents 10A along a, and 20A along $-a_z$ find
 (i) H_y at (-3, 4, 5)
 (ii) H_z at (-3, 4, 5)

(6marks)

(5marks)

(7marks)

(7marks)

QUESTION FOUR

- State three differences between Dynamic and static electromagnetic field
- Given

(9marks)

$$V_{emf} = -d\psi/dt$$

(6marks)

- An area of 0.65m^2 in the $Z = 0$ plane is enclosed by a filamentary conductor. Find the induced voltage, given that

$$B = 0.05 \cos 10^3 t \left[\frac{a_y + a_z}{\sqrt{2}} \right] \text{ Tesla}$$

(10marks)

QUESTION FIVE

- State three fundamental characteristic of electromagnetic wave
- i. Define lossy Dielectric.
 ii. Briefly explain pointing theorem
- In a lossless medium for which $\eta = 60\pi$, $\mu_r = 1$ and
 $H = -0.1 \cos(\omega t - Z) a_x + 0.5 \sin(\omega t - Z) a_y \text{ A/m}$.

(6marks)

(4marks)

(5marks)

Calculate,

i) ϵ_r

(3marks)

ii) ω

(2marks)

iii) E

(5marks)

QUESTION SIX

- Briefly explain the phenomenon of reflection of a plane wave at normal incidence.
- With the aid of diagram, briefly explain the Transverse Electromagnetic wave (TEM)
- Given a uniform plane wave in air as
 $E_i = 40 \cos(\omega t - \beta Z) a_x + 30 \sin(\omega t - \beta Z) a_y \text{ V/m}$.
 Find H_i

(5marks)

(5marks)

(15marks)